Econ 10 Summary Paper

Introduction and Analysis:

Throughout the course of the semester, taking intro to statistical inference, has allowed for the opportunity to gain a better understanding of how we can use statistics to make educated predictions using sample data. At the start of the planning stage our group came up with the idea to test the statistically significant relationship between family size and household income. With this question in mind we came up with the hypothesis that larger households are associated with a higher income on average. However, before we can begin our whole project we had to come up with multiple control variables that might have some effect on household income. With this, we came up with a variable key to show what each variable we used in Stata means.

Variables:

HHINCOME: Total household income (pre-tax)

FAMSIZE: Family size (number of household members)

UHRSWORK: Usual hours worked per week

AGE: Age of the respondent

SEX: Sex (male or female)

MARST: Marital status RACE: Race (recoded as White vs. Non-White)

EMPSTAT: Employment status (employed vs. not employed)

EDUCD: Educational attainment (detailed categories)

CLASSWKR: Class of worker

HHWT: Household weight

Linear Model:

$$HHINCOME = \alpha + \beta1(FAMSIZE) + \beta2(AGE) + \beta3(RACE) + \beta4(SEX) + \beta5(EMPSTAT) + \beta6(UHRSWORK) + \beta7(EDUCD) + \beta8(CLASSWKR)$$

Data and Variable Construction:

The dataset collected and used in this study contains approximately 3 million observations obtained from IPUMS USA. Data cleaning and variable construction were performed using Python (Jupyter Notebook). Specifically, the dependent variable is total household income (HHINCOME), defined as the total income before taxes earned by all individuals living in the household during the previous year. Family size (FAMSIZE) is the treatment variable. This is measured as the total number of people living in each household. The control variables include age, race, sex, employment status, hours worked per week, educational attainment, and worker class.

The statistics indicate a household income of approximately $128,726, taking into account that it is an average-sized family, specifically of 3 people, and also considering that their total working hours are 38 hours per week.

Descriptive Statistics:

Table 1: Descriptive statistics for continuous variables

	Total Household Income	Family Size	Age	Hours Worked / Week
N (sample size	3228708.00	3405809.00	3405809.00	1758960.00
Mean	128725.74	3.00	43.11	37.69
median	95600.00	3.00	44.00	40.00
Std. Dev.	130647.68	1.72	23.97	13.16
Minimum	0.00	1.00	0.00	1.00
Maximum	3002000.00	20.00	96.00	99.00

```
Table 1: Descriptive Statistics for Continuous Variables
           Total Household Income ($)  Family Size         Age  \
N                        3228708.00   3405809.00  3405809.00
Mean                      128725.74         3.00       43.11
Median                     95600.00         3.00       44.00
Std. Dev.                 130647.68         1.72       23.97
Minimum                        0.00         1.00        0.00
Maximum                  3002000.00        20.00       96.00

           Hours Worked / Week
N                  1758960.00
Mean                    37.69
Median                  40.00
Std. Dev.               13.16
Minimum                  1.00
Maximum                 99.00
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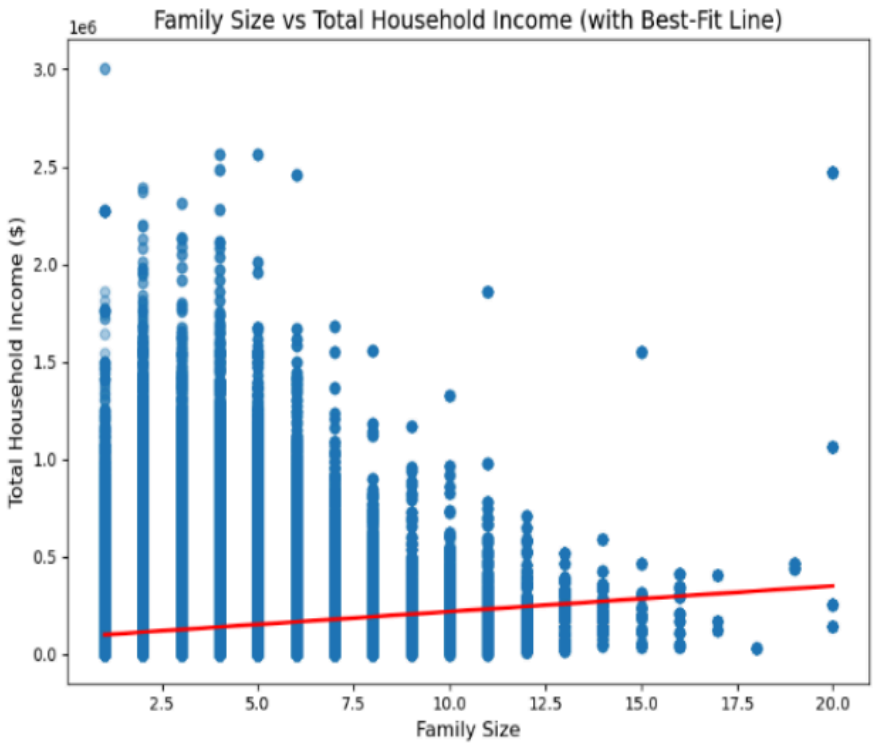
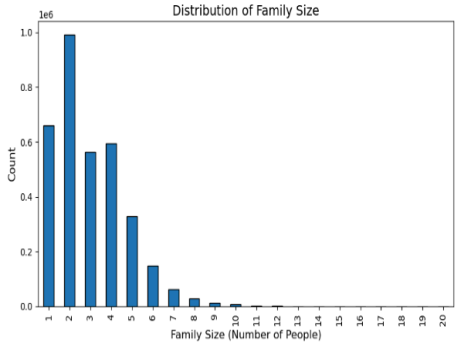
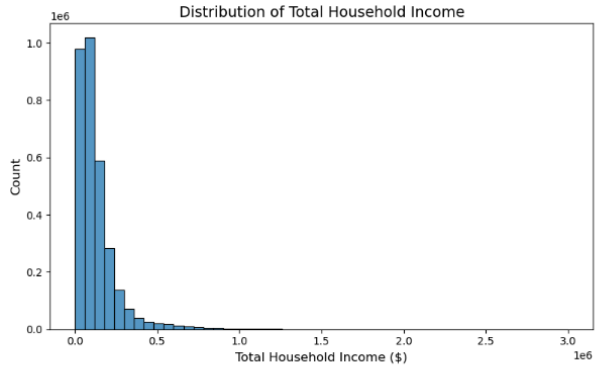
The income earned by the family varies, meaning it can range from that of one or two people to approximately twenty. Table 1 is basically summarizing the specific key continuous variables used in the analysis. The mean total household income is about $128,726; however, the median income is $95,000 meaning that there is a right-skew distribution because of a small number of high-income households. The average family size is 3 people individually which includes both parents, meaning more often than not just one kid. Most houses have between one and four family members, the wide range in income means every household earns very different amounts of money. Some earn very little, some earn extremely high incomes, some extremely low. Most though are in between because family size alone does not and can not explain income. Meaning it is important to account for more demographic and employment circumstances

Results and Analysis:



The scatter plot of family size and household income has a clear linear trend. This suggests that a modest positive correlation is happening between family size and household income. At the same time, income varies and is dispersed quite widely across households with similar family sizes, meaning that family size is one of many factors that correlate a household's total income.





The histograms of household income and family size show a right-skewed distribution of income and a concentration of households with small family sizes. The categorical distributions show that these are approximately balanced by the sex of each person in the household, indicating that almost half of the households are married, and most individuals work the expected number of hours.

Table 2: Frequency Table for Categorical variables

Category	Percentage
sex	
- Male	48.7%
- Female	51.3%
Martial status	
- Married, spouse present	49.2%
- Never married/single	29.7%
- Divorced	10.7%
- Widdowed	6.6%
- Married, spouse absent	2.3%
- Separated	1.4%
Race (white/non-white)	
- White	68.3%
- Non-White	31.7%
Employment status	
- Employed	56.1%
- Not employed	43.9%

Difference of means test

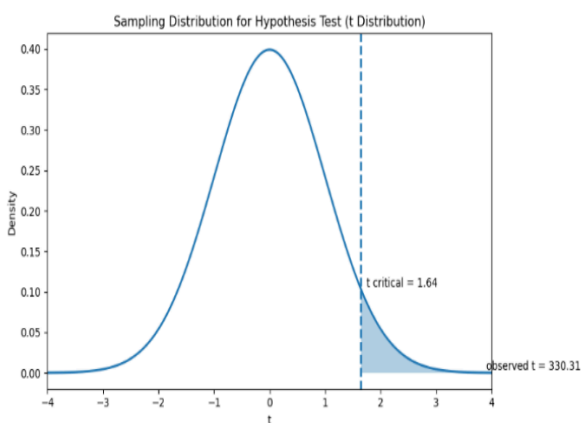
To assess whether household income differs between large and small families, we divided our sample based on the median family size. We observed that households with fewer family members (1 to 3 people) have an average annual income of approximately $103,350, while families with more members (3 or more people) have an average income of $150,036.

T-test of income and Family Size

	Low Family Size (< or = to 3)	High Family Size (> 3)
Sample Size (N)	**1,473,802**	**1,754,906**
Mean Annual Income	**10,3350.24**	**150036.56**

A T-test was performed to compare the average incomes of the two families.

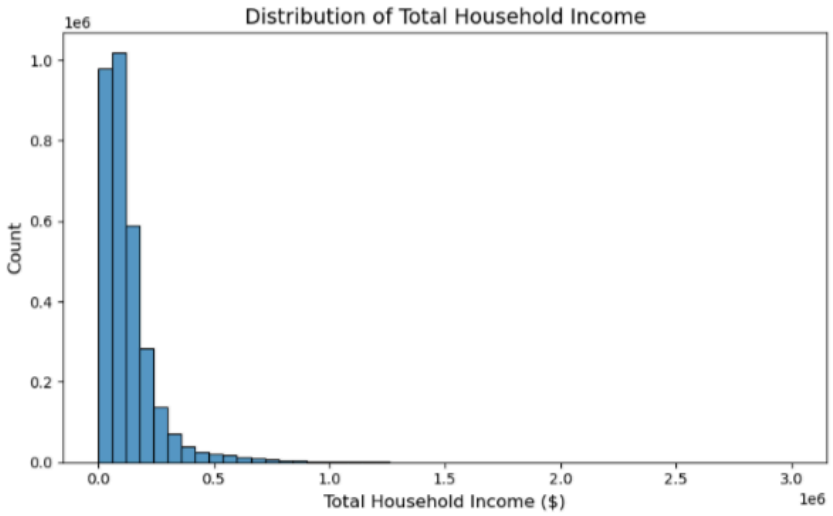
The results show that the T-test exceeds the critical value, indicating that the P-value is less than 0.05%, which leads to the rejection of the null hypothesis.

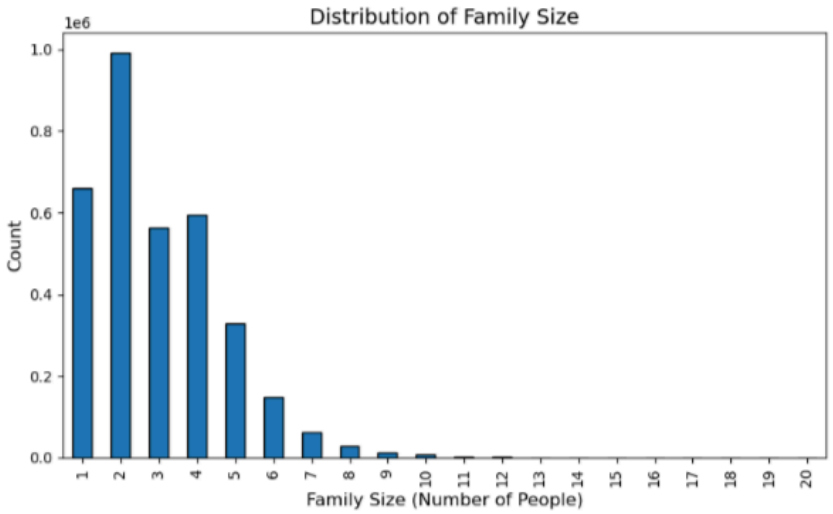


Having the answers to these tests provides evidence that income can significantly influence each household, regardless of the number of family members.

Regression analysis

The simple linear regression of household income on family size indicates a strong positive relationship, with an increase of approximately $15,000 for each additional person in the household. This suggests that households with more members tend to report higher total income to the government, but the model does not control for relevant confounding factors. The scatter plots of family size versus income generated from their jobs provide evidence of significant income dispersion across different family sizes.





To obtain a more precise estimate of the effect that family size can have, it is necessary to perform a multiple regression model that can control for several different demographic and labor variables. This model yields a negative coefficient, specifically a statistically significant negative relationship between family size and income ($\beta \approx -0.1179$, $p < 0.001$). This result is highly significant because it indicates that, after controlling for age, education, marital status, employment status, and hours worked, households with more family members tend to have lower income levels.

Model strength

The adjusted R^2 value is approximately 0.167, which indicates that the model can explain about 17% of the variation in household income. However, the statistics on the effect of family size are strong, even though the overall explanatory power of the model is moderate. This suggests that household income is significantly influenced by other important factors that are not immediately apparent, such as regional wage differences, occupation type, and the cost of living.

Discussion

Each result obtained reveals the relationship between family income and household size. Without controlling for family size, there is a positive correlation between household size and income, because larger households are more likely to include extra income earners or have all members contributing to the household income. However, when demographic and employment characteristics are controlled for, this correlation weakens and tends to reverse. Similarly, it is observed that families with more members face greater pressure to cope with financial crises, as

they incur higher personal care costs. These findings highlight a key aspect of economic theory, which states that income does not increase significantly in proportion to household size.

Conclusion

This entire study on family income and family size reveals a statistically significant relationship. We found that, after controlling for demographic and employment characteristics, households with larger families are associated with lower income levels. While family size alone doesn't fully explain the variation in income, it remains a very important determinant. These findings have significant implications for the design of public policies, particularly in tax credit assistance programs and poverty measures that take into account the economic realities of households of all sizes.